

Short, clear description of database

The schema is separated into 7 tables: candidates, departments, skills, positions, interviews, candidate_skills, and position_skills_req

The latter 2 tables are used to create foreign keys between candidates & skills, and positions & skills, respectively.

candidates: All the main information about each candidate, except for their skills

departments: All the main information about each department

skills: List of all possible skills

positions: Information about each available position

interviews: All the main information about each interview, except for the skills required

candidate_skills: Each row is one (of possibly many) of a candidate's skills

position_skills_req: Each row is one (of possibly many) of a position's required skills

Discussion of assumptions / additions made

Assumptions

- I assumed that the "skills" part of the tables didn't have to be a column in the table, as there could be any number ($n > 0$) of skills for each candidate or position. Because of this, I made 2 new tables to link the set of all skills to each candidate or position.
- I assumed phone numbers would be in the Irish mobile format: **0XX_XXX_XXXX** and so instead of using INT as the data type, I used VARCHAR(12) to allow for the spaces after the first 3 digits, and second 3 digits. I also needed to do this because the INT type was correcting any phone number **0ABCDEFGHI** to be **8BCDEFGHI**, dropping the leading zero

Additions

- One of the columns in "interviews" is **position_offered?** And I wanted to make it binary. (i.e. yes/no or true/false etc.) This is not a supported data type in SQL, so I used a TINYINT(1) to allow it to be any 1-digit number. Then I added a constraint so that only 0 and 1 were legal values

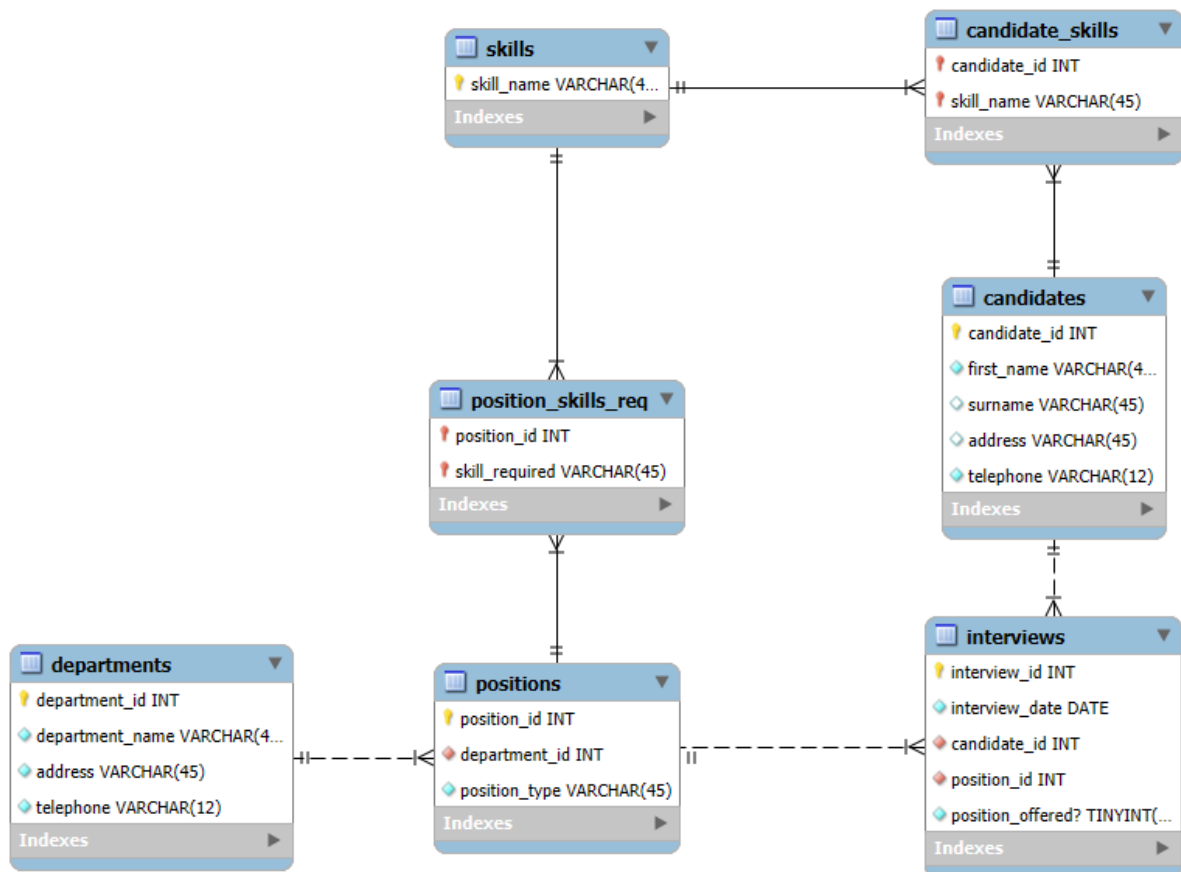
```
alter table interviews
add column `position_offered?` tinyint(1) not null,
add constraint chk_boolean_value check (`position_offered?` in (0, 1));
```

- I selected the "auto-increment" option for the primary keys of **candidates**, **departments**, and **skills**. This meant that in the stored procedure for adding a new row, the user doesn't worry about identification numbers, and they are generated by SQL.

Discussion of reaction policies used and why

- Foreign keys between all appropriate tables to maintain referential integrity
- Constraints on the position_offered? column of the interviews table
- Having this attribute default to be 0 (but allow for input in the saved procedure)
- Used the DATE data type for any dates in the database
- Using the auto-increment feature to generate id numbers for tables

ER Diagram of the database



Operating System used

Windows 11